Chapter 8
Functions

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What we will learn

- Read a function's *domain*, *codomain*, and rule together.
- Prove or refute injectivity, surjectivity, and bijectivity.
- Use composition and inverses to organize proofs.
- Work with ordered pairs and a bijection $\mathbb{N}^2 \to \mathbb{N}$.

A function includes its types

$f : X \to Y$ assigns exactly one element $f(x) \in Y$ to every $x \in X$. Its *image* is the set of outputs actually attained; its codomain is all of Y.

The rules $n \mapsto 2n$ on $\mathbb{Z}$ and $x \mapsto 2x$ on $\mathbb{R}$ have different surjectivity behavior.

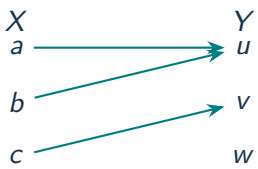
Source: Heather Macbeth, *The Mechanics of Proof*, Section 8.1.

8.1: Injectivity asks whether outputs identify inputs

Definition

$f : X \to Y$ is *injective* when

$$\forall x_1, x_2 \in X, \qquad f(x_1) = f(x_2) \implies x_1 = x_2.$$



This function is not injective: $a \neq b$ but $f(a) = f(b) = u$.

A counterexample must give *two distinct inputs with the same output*. An unused output concerns surjectivity instead.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.1.2; Examples 8.1.8–8.1.11.

Two complete injectivity arguments

Translation: $q : \mathbb{R} \to \mathbb{R}$, $q(x) = x + 3$

Let $x_1, x_2 \in \mathbb{R}$ and assume $q(x_1) = q(x_2)$. Then

$$x_1 + 3 = x_2 + 3 \quad \Longrightarrow \quad x_1 = x_2$$

by subtracting 3. These inputs were arbitrary, so q is injective.

Squaring: $s : \mathbb{R} \to \mathbb{R}$, $s(x) = x^2$

Choose $x_1 = -1$ and $x_2 = 1$. We have

$$s(-1) = 1 = s(1), \qquad -1 \neq 1.$$

Thus s fails the defining implication and is not injective.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.1.3–8.1.4.

Surjectivity asks whether every target is reached

Definition

$f : X \to Y$ is *surjective* when $\forall y \in Y,\ \exists x \in X,\ f(x) = y$.

Example. Let $f : \mathbb{Q} \to \mathbb{Q}$ satisfy $f(x) = 3x + 2$. Given any $y \in \mathbb{Q}$, choose $x = (y - 2)/3 \in \mathbb{Q}$. Then

$$f(x) = 3\frac{y-2}{3} + 2 = y.$$

The witness depends on the arbitrary target y, so this proves surjectivity.

Failure. Squaring $\mathbb{R} \to \mathbb{R}$ misses -1, since $x^2 \geq 0$ for every real x. A failed search for a witness is insufficient; we proved no witness can exist.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.1.5; Examples 8.1.6–8.1.7.

From quantifiers to Lean proof steps

```
example : Function.Injective
    (fun x : Real => x + 3) := by
  intro x1 x2 h
  dsimp at h
  addarith [h]

example : Function.Surjective
    (fun x : Rat => 3 * x + 2) := by
  intro y
  use (y - 2) / 3
  dsimp
  ring
```

`intro` selects arbitrary inputs or a target; `use` supplies a preimage. For negative claims, unfold the property and use `push_neg`.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.1.3 and 8.1.6; illustrative textbook-environment code.

Practice 8.1: Give a proof or a specific obstruction

1. Is $x \mapsto 2x$ surjective as a function $\mathbb{R} \to \mathbb{R}$? As a function $\mathbb{Z} \to \mathbb{Z}$?
2. Let $X = \{a, b, c\}$ and $Y = \{u, v\}$, with

x	a	b	c
$f(x)$	u	v	u

   Is $f : X \to Y$ injective? Is it surjective?

For each positive answer, account for every relevant input or target. For each negative answer, state the exact witnesses.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 8.1.13; original finite-table exercise.

Practice 8.1: Solution ideas

1. Over $\mathbb{R}$, target y has preimage $y/2$, because $2(y/2) = y$. Over $\mathbb{Z}$, target 1 has no preimage: $2n$ is even for every $n \in \mathbb{Z}$.
2. The finite function is not injective: $a \neq c$ and $f(a) = f(c) = u$. It is surjective: u has preimage a, and v has preimage b. These are all the targets in Y.

Proof structure in Lean

Surjectivity over a finite type: `intro y`, then `cases y`, then a witness in each case. The book's `exhaust` handles the resulting concrete finite statements.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.1.9 and 8.1.11; Exercises 8.1.13.

8.2: Bijectivity combines two obligations

Definition

$f : X \to Y$ is *bijective* when it is both injective and surjective.

Example. Let $p : \mathbb{R} \to \mathbb{R}$, $p(x) = 2x - 5$.

1. **Injectivity.** If $p(x_1) = p(x_2)$, then $2x_1 - 5 = 2x_2 - 5$. Adding 5 and dividing by 2 gives $x_1 = x_2$.
2. **Surjectivity.** Given $y \in \mathbb{R}$, let $x = (y + 5)/2$. Then

$$p(x) = 2\frac{y+5}{2} - 5 = y.$$

Both parts hold with domain and codomain $\mathbb{R}$, so p is bijective.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.2.1; Example 8.2.2.

A bijection gives exactly one preimage

Theorem

For $f : X \to Y$, f is bijective if and only if

$$\forall y \in Y, \quad \exists! x \in X \text{ such that } f(x) = y.$$

Forward. Surjectivity gives a preimage x of y. If $f(x') = y$ also, then $f(x') = f(x)$; injectivity gives $x' = x$.

Backward. Existence gives surjectivity. To prove injectivity, assume $f(x_1) = f(x_2)$ and call their common output y. Both x_1 and x_2 are preimages of y, so uniqueness gives $x_1 = x_2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.2.5.

Lean: Split the bijection proof

```
example : Function.Bijective
    (fun x : Real => 2 * x - 5) := by
  constructor
  case left =>
    intro x1 x2 h
    dsimp at h
    addarith [h]
  case right =>
    intro y
    use (y + 5) / 2
    dsimp
    ring
```

`constructor` separates injectivity from surjectivity. To refute bijectivity, it is enough to refute either part.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.2.2–8.2.4; illustrative textbook-environment code.

Practice 8.2: Two ways a bijection can fail

1. For $s : \mathbb{N} \to \mathbb{N}$, $s(n) = n + 1$, prove injectivity and disprove bijectivity.
2. For $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^2 + 2x$, disprove bijectivity using two inputs.

Plan before calculating

For the first function, look for a target below every output. For the second, factor $x^2 + 2x = x(x + 2)$ and compare its zeros.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.2.7; Exercises 8.2.8.

Practice 8.2: Solution ideas

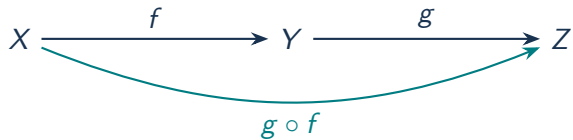
1. If $n_1 + 1 = n_2 + 1$, cancellation gives $n_1 = n_2$, so s is injective. But $s(n) = n + 1 > 0$ for all $n \in \mathbb{N}$. Thus 0 has no preimage, and s is not bijective.
2. We have $h(0) = 0$ and $h(-2) = 4 - 4 = 0$, whereas $0 \neq -2$. Thus h is not injective, which already disproves bijectivity.

Finite versus infinite domains

On the same finite type, an injective self-map is bijective (the book checks small examples). The successor function shows that the analogous statement for $\mathbb{N}$ is false.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.2.6–8.2.7; Exercises 8.2.8.

8.3: Composition follows the types



For $f : X \to Y$ and $g : Y \to Z$, define $(g \circ f)(x) = g(f(x))$. The identity $\mathrm{id}_X : X \to X$ satisfies $\mathrm{id}_X(x) = x$.

If $f(x) = x + 3$ and $g(x) = 2x$ on $\mathbb{R}$, then

$$(g \circ f)(x) = 2(x + 3) = 2x + 6, \qquad (f \circ g)(x) = 2x + 3.$$

Order matters. Equality of functions means equality at *every* input.

Source: Heather Macbeth, *The Mechanics of Proof*, Definitions 8.3.1 and 8.3.3; Example 8.3.2.

Lean: Extensionality changes the kind of goal

```
def f (x : Real) : Real := x + 3
def g (x : Real) : Real := 2 * x
def h (x : Real) : Real := 2 * x + 6

example : (fun x => g (f x)) = h := by
  ext x
  dsimp [f, g, h]
  ring
```

Before `ext x`	Equality of two functions
After `ext x`	Equality of their values at an arbitrary x
After `dsimp`	$2(x+3) = 2x+6$

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.3.2; illustrative textbook-environment code.

An inverse must undo the function in both directions

Definition

For $f : X \to Y$, a function $g : Y \to X$ is an *inverse* if

$$g \circ f = \mathrm{id}_X \qquad \text{and} \qquad f \circ g = \mathrm{id}_Y .$$

Equivalently: $g(f(x)) = x$ for every $x \in X$, and $f(g(y)) = y$ for every $y \in Y$.

For $s : \mathbb{R} \to \mathbb{R}$, $s(x) = 5 - x$, we obtain

$$s(s(x)) = 5 - (5 - x) = x \qquad (x \in \mathbb{R}).$$

Thus s is its own inverse. Since the two functions are the same here, the one calculation establishes both required identities.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.3.5; Example 8.3.4.

A function is bijective exactly when it has an inverse

Suppose $f : X \to Y$ is bijective. For each y, let $g(y)$ be its unique preimage. Then $f(g(y)) = y$. For any x,

$$f(g(f(x))) = f(x) \quad \Longrightarrow \quad g(f(x)) = x$$

by injectivity. Hence g is an inverse.

Suppose $g : Y \to X$ is an inverse. If $f(x_1) = f(x_2)$, then

$$x_1 = g(f(x_1)) = g(f(x_2)) = x_2.$$

For any $y \in Y$, $g(y) \in X$ is a preimage because $f(g(y)) = y$. Thus f is injective and surjective.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.3.7–8.3.9; Lean uses choose to select preimages.

Practice 8.3: Build a function proof from smaller ones

1. Let $u : \mathbb{R} \to \mathbb{R}$ satisfy $u(x) = 5x + 1$. Find an inverse and verify both compositions.
2. Let $f : X \to Y$ and $g : Y \to Z$ be injective. Prove $g \circ f$ is injective.
3. With the same types, assume instead that both functions are surjective. Prove $g \circ f$ is surjective.

For composition, write the intermediate value in Y explicitly. It is where the two hypotheses meet.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 8.3.10.

Practice 8.3: Solution ideas

1. Set $v(y) = (y-1)/5$. Then
$$v(u(x)) = \frac{5x+1-1}{5} = x, \qquad u(v(y)) = 5\frac{y-1}{5} + 1 = y.$$
2. If $g(f(x_1)) = g(f(x_2))$, injectivity of g gives $f(x_1) = f(x_2)$; injectivity of f then gives $x_1 = x_2$.
3. Given $z \in Z$, choose $y \in Y$ with $g(y) = z$, then choose $x \in X$ with $f(x) = y$. Consequently $g(f(x)) = g(y) = z$.

In Lean, the second proof applies two injectivity hypotheses; the third extracts two existential witnesses and then uses x.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 8.3.10.

8.4: Ordered pairs keep both coordinates

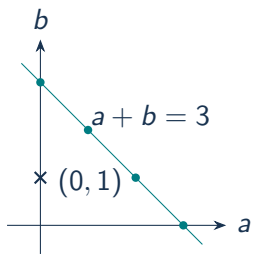
The product $X \times Y$ consists of pairs (x, y) with $x \in X$ and $y \in Y$.

$$(x_1, y_1) = (x_2, y_2) \iff x_1 = x_2 \ \wedge \ y_1 = y_2.$$

Let $q : \mathbb{Z} \to \mathbb{Z}^2$, $q(m) = (m + 1, 2 - m)$.

- Equal outputs give $m_1 + 1 = m_2 + 1$ from the first coordinate, hence $m_1 = m_2$.
- Every output (a, b) satisfies $a + b = 3$. Since $0 + 1 \neq 3$, $(0, 1)$ has no preimage.

So q is injective but not surjective.



Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.1.

Solve for an inverse, then verify it

Let $T : \mathbb{Z}^2 \to \mathbb{Z}^2$ satisfy $T(m, n) = (m + n, m + 2n)$. If $(a, b) = T(m, n)$, subtracting the two equations gives

$$n = b - a, \qquad m = a - n = 2a - b.$$

Thus define $U : \mathbb{Z}^2 \to \mathbb{Z}^2$ by $U(a, b) = (2a - b, b - a)$. Both components are integers. Now check:

$$\begin{aligned} U(T(m, n)) &= (2(m + n) - (m + 2n),\ (m + 2n) - (m + n)) \\ &= (m, n), \\ T(U(a, b)) &= ((2a - b) + (b - a),\ (2a - b) + 2(b - a)) \\ &= (a, b). \end{aligned}$$

U is an inverse, so T is bijective.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.2.

Lean: Function equality and coordinate equality

```
def T (p : Prod Int Int) : Prod Int Int :=
  (p.1 + p.2, p.1 + 2 * p.2)
def U (p : Prod Int Int) : Prod Int Int :=
  (2 * p.1 - p.2, p.2 - p.1)

example : And ((fun p => U (T p)) = id)
              ((fun p => T (U p)) = id) := by
  constructor <;> ext p <;> dsimp [T, U] <;> ring
```

- `Prod Int Int` is the ASCII spelling of $\mathbb{Z} \times \mathbb{Z}$.
- `p.1` and `p.2` access the two coordinates.
- `<;>` sends the following tactic to every generated goal.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.2; illustrative textbook-environment code.

Enumerating pairs of natural numbers

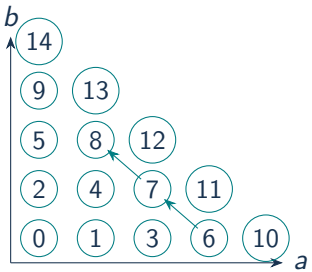
Theorem

There exists a bijection $\mathbb{N}^2 \to \mathbb{N}$.

Define $A_0 = 0$ and $A_{s+1} = A_s + s + 1$. The first values are $0, 1, 3, 6, 10, \ldots$.

$$p(a, b) = A_{a+b} + b.$$

On diagonal $a + b = s$, the second coordinate b runs from 0 through s. Thus this diagonal gets the labels $A_s, \ldots, A_s + s$.



Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.9.

Why the diagonal enumeration is a bijection

Diagonal s receives exactly the integer interval

$$I_s = \{A_s, A_s + 1, \ldots, A_s + s\} = \{A_s, \ldots, A_{s+1} - 1\}.$$

Injectivity. Different diagonals have disjoint intervals, since the intervals are consecutive and A_s strictly increases. Equal labels therefore have the same s. Within that diagonal, $A_s + b$ determines b, and then $a = s - b$.

Surjectivity. $I_0 = \{0\}$ and each interval starts just after the previous one ends. They cover $\mathbb{N}$: inductively, the successor of a covered number stays in its interval or is the first number of the next interval. If $k \in I_s$, take $b = k - A_s$ and $a = s - b$. Then $0 \leq b \leq s$ and $p(a, b) = k$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.9; alternative written proof of the same construction.

Practice 8.4: Check that the inverse stays in the domain

Consider $H(m, n) = (m + n, m - n)$.

1. Prove $H : \mathbb{R}^2 \to \mathbb{R}^2$ is bijective by constructing an inverse.
2. Decide whether the same rule gives a bijection $\mathbb{Z}^2 \to \mathbb{Z}^2$. Test the target $(0, 1)$.
3. Further practice: prove $(m, n) \mapsto 5m + 8n$ is surjective from $\mathbb{Z}^2$ to $\mathbb{Z}$. Use $5(-3) + 8(2) = 1$.

Solving equations over $\mathbb{R}$ does not ensure the resulting preimage has integer coordinates.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.4.3 and 8.4.6.

Practice 8.4: Solution ideas

1. Over $\mathbb{R}$, set $K(a,b) = ((a+b)/2, (a-b)/2)$. Then
$$K(H(m,n)) = (m,n), \qquad H(K(a,b)) = \left(\tfrac{a+b+a-b}{2}, \tfrac{a+b-a+b}{2}\right) = (a,b).$$
2. Over $\mathbb{Z}$, $H(m,n) = (0,1)$ would give $m+n=0$ and $m-n=1$. Adding gives $2m=1$, impossible for an integer m. Hence H is not surjective.
3. For any $z \in \mathbb{Z}$, the pair $(-3z, 2z)$ is an integer preimage: $5(-3z)+8(2z) = z$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.4.3 and 8.4.6.

Chapter 8 review: Choose the obligation first

Goal	Useful first move
Injective	Assume equal outputs; recover equal inputs.
Not injective	Exhibit distinct inputs with equal outputs.
Surjective	Fix a target; construct a preimage in the domain.
Not surjective	Exhibit a target and rule out every preimage.
Bijective	Prove both properties, or verify an inverse.
Equal functions	Use an arbitrary input; in Lean, `ext`.
Equal pairs	Compare the corresponding coordinates.

Exit check. Explain exactly why division by 2 is harmless over $\mathbb{R}$ but can obstruct a claimed inverse over $\mathbb{Z}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Sections 8.1–8.4.

Reading and code context

Heather Macbeth, *The Mechanics of Proof*, Chapter 8:

- 8.1 Injectivity and surjectivity
- 8.2 Bijectivity
- 8.3 Composition of functions
- 8.4 Product types

`https://hrmacbeth.github.io/math2001/08_Functions.html`

These notes reorganize selected examples and supply new explanations and diagrams. Lean listings illustrate the textbook's teaching environment, including `addarith`, `numbers`, and `exhaust`. ASCII type names such as `Real` and `Prod Int Int` spell out the corresponding mathematical types.